

# JOOYOUNG YOO

## CONTACT INFORMATION

jooyoung.yoo@emory.edu

<https://jooyoung01.github.io>

## EDUCATION

### Emory University

*Ph.D. in Computer Science (Advisor: Dr. Joon-Seok Kim)*

PRESENT

### University of Southern California

*M.S. in Spatial Data Science (Advisor: Dr. John Wilson, and Dr. Yi Qi)*

2025

### Myongji University

*M.S. in Data Technology (Advisor: Dr. Daewon Kim)*

2020

### Myongji University

*B.E. in Information and Communication Engineering*

2016

## PUBLICATIONS

**Jooyoung Yoo**, Yi Qi, Isaac Ashe-McNalley, Beau MacDonald, and John P. Wilson (2026). “An Open and Transferable Deep Learning Framework for Mapping Urban Tree Canopy Using NAIP Imagery” *Remote Sensing* 18, no. 12: 1899.

Joon-Seok Kim, **Jooyoung Yoo**, Su Yeon Han, Andreas Züfle (2025), “An Agile Design of Activity-based Mobility Intervention using Large Language Models” *The 8th ACM SIGSPATIAL International Workshop on Geospatial Simulation (GeoSIM '25)*, Nov.3, 2025

Su Yeon Han, Yubin Lee, Jooyoung Yoo, Jeon-Young Kang, Jinwoo Park, Soe W. Myint, Eunsang Cho, Xin Gu, and Joon-Seok Kim. “Spatial disparities in fire shelter accessibility: Capacity challenges in the palisades and eaton fires.” arXiv preprint arXiv:2506.06803 (2025).

Su Yeon Han, **Jooyoung Yoo**, Alexander Michels, Shaowen Wang, Joon-Seok Kim (2025), “Vulnerable Neighborhood Explorer (VNE): An Open-Source Visual Analytics Tool for Exploring Social Vulnerability to Disasters across Different Neighborhoods” *SoftwareX*, 31, p.102233.

Jingu Kim, **Jooyoung Yoo**, Eun Young Kang (2025), “Exploring Social Interdependence and Peer Relationships of University Students: A Longitudinal Social Network Analysis.” *Journal of Educational Innovation Research*, 35, no. 2: 25–46.

Siqin Wang, **Jooyoung Yoo**, Wenhui Cai, Fan Yang, Xiao Huang, Qian Chayn Sun, Shaokun Lyu (2025). “Reducing the social inequity of neighborhood visual environment in Los Angeles through computer vision and multi-model machine learning.” *Sustainable Cities and Society*, 119, p.106062.

**Jooyoung Yoo**, Reem Emad Shtaiwi, Mohammad Yasin, Dweep Trivedi, Abdullah Alfarrarjeh, Amani Abu Jabal, Seon Ho Kim (2024), “Towards real-world deployment of deep learning solutions for global road damage detection and classification,” *IEEE Big Data, Big Data Cup Challenge (ORDDC'2024)*, **awarded the Bronze Prize in an international challenge**, pp. 8485-8494. Dec.17, 2024

**Jooyoung Yoo**, Abdullah Alfarrarjeh, Krish Sukhani, Seon Ho Kim, and Cyrus Shahabi (2024). “STVD: Synchronized Truck Video Dataset with RGB and IR cameras for continuous truck traffic monitoring,” *IEEE International Workshop on AI and Computer Science*, Dec.9, 2024

**Jooyoung Yoo**, Seon Ho Kim, Krish Sukhani, Min Sang Yoo, and Cyrus Shahabi (2024). “Truck detection and counting in low-light condition: do we need infrared camera?,” *In IEEE International Conference on Big Data and Smart Computing*, Feb.15, 2024

**Jooyoung Yoo**, and Daewon Kim (2020). “Development of a new pedestrian avoidance algorithm considering a social distance for social robots,” *Journal of Broadcast Engineering* 25.5 : 734-741.

**Jooyoung Yoo**, and Jin Lee Park (2020). “Examining the socialization of grit and the effects of peer community and teacher closeness using longitudinal social network analysis,” *Journal of Learner-*

**Jooyoung Yoo**, Joomin Kim, Sungsik Yun, and Daewon Kim (2013). “Development of a scenario-based work distribution function for teleoperation under multi-user and multi-robot environments,” *IEEE International Symposium on Robotics 2013*

RESEARCH  
EXPERIENCE

**Urban World Model and Multimodal Spatial Representation Project** JUL. 2026 – PRESENT  
*Ph.D. Student (Advisor: Dr. Joon-seok Kim)*

- Developing an urban world model that learns unified spatial representations from heterogeneous geospatial data, including aerial imagery, LiDAR, OpenStreetMap, and urban attributes.
- Investigating multimodal embedding and fusion methods to align different representations of the same urban environment into a shared latent space.
- Exploring generative world models and 3D simulation environments for representing and simulating real-world urban environments.

**Individual Tree Detection, Segmentation for Urban Greening Project** MAY. 2024 – AUG. 2025  
*Full-time Student Researcher (Advisor: Dr. John Wilson, and Dr. Yi Qi)*

- Developed detection and segmentation models using aerial imagery to identify and segment individual trees.
- Designed a training and prediction workflow for LA city administrators, enabling them to count trees and calculate canopy areas using GIS, without the need for coding skills.

**Spatial Pattern Mapping using Street View: Human Perception Project** MAR. 2024 – SEP. 2024  
*Volunteer Student Researcher (Advisor: Dr. Siqin Wang)*

- Conducted data collection at 50-meter intervals across Los Angeles County using GIS software, Python libraries, and 360-degree imagery from Google and Mapillary Street View APIs.
- Applied machine learning methods, including segmentation and classification, to analyze Street View imagery for sentiment analysis of various locations.
- Visualized spatial patterns of six human emotions (Beautiful, Wealthy, Livable, Safe, Boring, Depressing) to map urban perception.

EMPLOYMENT

**Spatial Data Lab (SDL) at Harvard University**, *Intern* 2024 - 2025

**Spatial Sciences Institute of USC**, *Database Analyst, Full-time Researcher* 2024 - 2025

**Keck School of Medicine of USC**, *Full-time Student Researcher* 2023 – 2024

**Integrated Media Systems Center of USC**, *Teaching Assistant, Student Researcher* 2023 – 2024

**Joongbu University**, *Web Developer* 2020 – 2022

**Techsphere**, *Computer Vision AI Software Engineer* 2020 – 2021

**Myongji University**, *Research Assistant, Teaching Assistant* 2018 – 2020

**Konkuk University**, *Web Developer* 2018

**Korean Dictionary**, *Co-founder* 2013 – 2017

<sup>†</sup>Non-profit Organization

AWARDS

Winner – 2026 Esri Innovation Program Students of the Year, Esri 2026

Bronze Prize – Optimized Road Damage Detection Challenge 2024, IEEE BigData 2024

Best Presenter – The 23rd KOCSEA Technical Symposium 2023

Best Presenter – USC CKIDS DataFest Fall 2022 2022

SKILLS

**DL Frameworks:** Pytorch, Keras, Tensorflow, Ultralytics

**Languages:** Python, R, C++, Java, JavaScript, SQL

**Other:** Git, Linux, Matlab, ArcGIS Pro, QGIS, SPSS, L<sup>A</sup>T<sub>E</sub>X – All professional proficiency or above